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# Leveraging LLM based Retrieval-Augmented Generation for Legal Knowledge Graph Completion

Jiang Lai[1], Zheng Conghui[1], Zhang Xiaohan[1], Sun Fuhui[2], Wang Xiaoyan[2], Pan Li[1,*]
[1] *Institute of Cyber Science and Technology, School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China*
[1] {future2020, chzheng, xhzhang1, panli}@sjtu.edu.cn
[2] *Information Technology Service Center of People's Court, Beijing, China*
[2] sunfh6732@163.com, 428163395@139.com

***Abstract*—Aiming at filling missing triple structures, Knowledge Graph Completion (KGC) is a crucial task in knowledge graph reasoning. Especially in the judicial domain, it plays a significant role not only in enhancing the completeness of legal knowledge graphs but also in improving the accuracy of legal supervisory inference. Due to extensive legal expertise, complex entity relationships, and strict requirements for interpretable reasoning, traditional methods, such as embedding-based methods and those that utilize pre-trained language models (e.g., BERT), often fail to delve into deep semantic information or overlook the interconnections among triples, which leads to significant shortcomings in generalizability and capabilities in practical applications. Motivated by the great generation power of Large Language Models (LLMs), we propose a legal knowledge graph completion model based on Retrieval Augmented Generation (RAG), which we have named RA-KG-LLM. This model effectively combines the generative capabilities of LLMs with retrieval-augmented technology to enhance the semantic information and interrelations mining within knowledge graphs. The retrieval framework within our model utilizes text embeddings and vector similarity matching to provide the LLM with relevant triples, while fine-tuning techniques are employed to infuse the semantic information from the knowledge graph into the LLM. Extensive experiments on five real datasets demonstrate the effectiveness and potentiality in the judicial field of the proposed model. Specifically, on the legal domain dataset Cail2022, it achieves better Hits@1 score in the relation prediction task than the state-of-the-art related works.**

***Index Terms*—Large Language Models, Judicial Knowledge Graph, Knowledge Graph Completion, Retrieval-Augmented Generation**

## I. INTRODUCTION

With the rapid development of information technology, the application of big data and artificial intelligence in the judicial domain has deepened. Among these, knowledge graphs, as a crucial technological means for connecting complex data points, play an increasingly vital role in the organization of legal information and the process of legal reasoning[1]. Knowledge graphs provide legal professionals with an intuitive and structured tool for data analysis by constructing graphical structures of entities and their relationships. Knowledge graph completion[2] (KGC) is an essential task paradigm in knowledge graph reasoning and plays an indispensable role in applications within the judicial domain, such as the enhancement of judicial knowledge graphs and legal supervisory inference.

In the judicial field, knowledge graph completion not only needs to handle vast and complex legal expertise but also must address the intricate relationships between entities and the demand for high-precision reasoning. These challenges have revealed limitations in traditional knowledge graph completion methods, such as those based on embeddings and pre-trained language models. For example, methods based on embeddings[3], although capable of capturing low-dimensional representations of entities and relationships, often lack the exploration of deeper semantic information. In contrast, methods based on pre-trained language models[4], despite leveraging rich linguistic information, frequently overlook the specific associations between entities, impacting the model's reasoning capabilities.

To address these issues, we propose an innovative knowledge graph completion model based on retrieval-augmented generation with large models, called RA-KG-LLM. This model combines the powerful text generation capabilities of large language models with the latest retrieval-augmented generation[5] (RAG) technology, optimizing the processing of semantic information and associations within knowledge graphs. By incorporating retrieval-augmented mechanisms, the model can consider a broader context during prediction generation, significantly improving the accuracy and reliability of completion tasks.

The main contributions of this paper are as follows:

- A knowledge graph completion method that integrates the generative capabilities of large models and a retrieval-augmented framework is proposed, effectively addressing the limitations of traditional methods in handling complex judicial domain data.
- Experimental validation was conducted on multiple datasets, including general and legal domains, demonstrating significant advantages of this method over existing technologies. Particularly, on the legal domain dataset Cail2022, this method achieved outstanding performance

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in relation prediction tasks, proving its effectiveness in practical judicial applications.

The organization of this paper is as follows: The second section will detail related work, including traditional methods for knowledge graph completion and recent approaches based on large models. The third section will delve into our method, including the model architecture and specific implementation of the retrieval-augmented mechanism. The fourth section will present the experimental setup and results across multiple datasets, comparing and analyzing the performance of our method against other existing technologies. Finally, the fifth chapter will summarize the entire paper and look forward to future research directions.

## II. Related Works

Traditionally, knowledge graph completion methods are primarily classified into two categories: embedding-based methods and methods based on pre-trained language models (PLMs). Embedding-based methods map entities and relationships into a continuous vector space and utilize these embeddings to predict missing triplet information. TransE[3], one of the earliest translation-based embedding models, assumes that the difference between the head entity vector and the tail entity vector should be close to the relationship vector. RotatE[6] introduced a rotation-based embedding method, representing relationships as rotational operations in complex space, which better captures the symmetry, antisymmetry, and transitivity between entities. DistMult[7] and ComplEx[8] introduced tensor decomposition, predicting relationships between entities through multidimensional vector multiplication. ConvE[9] is a neural network-based method that learns embeddings for entities and relationships through convolutional neural networks. These methods effectively utilize the structural information of knowledge graphs through well-designed scoring functions and self-supervised learning strategies, but they often overlook the rich semantic information contained within entities and relationships.

Methods based on pre-trained language models treat knowledge graph completion as a text-based task, fine-tuning pre-trained models to understand textual descriptions of entities and relationships, thereby achieving knowledge graph completion. KG-BERT[4] was the first to model KGC as a binary text classification task using a pre-trained language model. MTL-KGC[10] and StAR[11] further expanded this approach by introducing more training tasks and complex encoding strategies to enhance model performance. KGT5[12] and KG-S2S[13], based on the encoder-decoder framework of the pre-trained language model T5[14], explored knowledge graph completion methods based on generative models. These methods leverage the strong semantic understanding capabilities of pre-trained language models, but their utilization of the associative information between knowledge graph triplets is limited.

With the emergence of large language models like GPT-3[15], some researchers have begun to explore applying these powerful models to knowledge graph completion tasks[16][17]. KG-LLAMA[18] fine-tunes large models to learn semantic knowledge of knowledge graphs and perform knowledge graph completion tasks through text generation, extending the methods based on pre-trained language models with similar limitations. KoPA[19] combines traditional embedding-based methods with large models, training a knowledge graph triple embedding model and fine-tuning the large model to understand the structural information represented by triplet embeddings, achieving good results in triple prediction tasks. However, the triple embedding model requires complete triplets as input, making this paradigm difficult to transfer to other tasks, such as entity prediction tasks.

In this context, we propose a knowledge graph entity and relationship completion model based on retrieval-augmented generation with large models, which we have named RA-KG-LLM. This model combines retrieval-augmented generation technology with the powerful generative capabilities of large models, while also introducing fine-tuning and few-shot prompting strategies, effectively enhancing the large model's understanding and completion capabilities for knowledge graphs, and demonstrating good adaptability in the judicial domain. Extensive experiments were conducted on multiple datasets, including four general domain datasets and one specialized legal domain dataset, validating the effectiveness of this method and its applicability in the judicial domain. Through this work, a new solution for knowledge graph completion tasks is provided, showcasing the potential of large models and retrieval-augmented generation technology in handling knowledge reasoning tasks in the judicial domain.

## III. KGC model Based on RAG with LLM

All symbols appearing in this paper are shown in Table I.

TABLE I
EXPLANATION OF SYMBOLS USED IN THE PAPER

| Symbol | Definition |
|---|---|
| $\mathcal{T}$ | A triple in the knowledge graph |
| $\mathbf{h}$ | The header Entity of the triple |
| $\mathbf{r}$ | The relation of the triple |
| $\mathbf{t}$ | The tail Entity of the triple |
| $\mathbf{v}$ | Vector representation of the triple |
| $\mathbf{f}$ | Pre-trained Embedding models |
| $\mathcal{D}$ | The vector database FAISS |
| $\mathcal{L}$ | Loss function of the embedding models |
| $\mathcal{P}$ | The prompt for large language models |
| $\mathbf{W}$ | Weight matrix of the large language model |
| $\mathbf{A}$ | Low-rank matrix of LoRA fine-tuning |
| $\mathbf{B}$ | Low-rank matrix of LoRA fine-tuning |

This section focuses on an emerging paradigm in large model technologies—Retrieval-Augmented Generation (RAG) and its application to the task of knowledge graph completion. This technique integrates both retrieval and generation mechanisms, aiming to enhance the generative capabilities of models by incorporating external knowledge. It is particularly suited for handling complex tasks that require the synthesis of external information during large model inference and generation, such as the completion of entities and relationships in knowledge graphs.

**Algorithm 1** Algorithmic Procedure of RA-KG-LLM

**// Stage 1. Preparation:**
**Input:** Set of triples $\mathcal{T}$ from the knowledge graph
**Output:** Vectorized triples stored in a vector database $\mathcal{D}$; Fine-tuned LLM
**for** each triple $(\mathbf{h}, \mathbf{r}, \mathbf{t})$ in $\mathcal{T}$ **do**
    $\mathbf{v} \leftarrow \mathbf{f}(\mathbf{h}, \mathbf{r}, \mathbf{t})$
    Store $\mathbf{v}$ in $\mathcal{D}$
**end for**
Initialize LoRA adapters for LLM
**for** each triple $(\mathbf{h}, \mathbf{r}, \mathbf{t})$ in $\mathcal{T}$ **do**
    Update $\mathbf{W}' = \mathbf{W} + \mathbf{AB}$ using $(\mathbf{h}, \mathbf{r}, \mathbf{t})$
**end for**
**// Stage 2. Generation:**
**Input:** Triple $\mathcal{T}'$ to be completed
**Output:** Completed triple
$\mathbf{v}' \leftarrow \mathbf{f}(\mathcal{T}')$
similar_triples $\leftarrow \mathcal{D}$.retrieve$(\mathbf{v}', \text{topk})$
$\mathcal{P} \leftarrow$ construct_prompt(similar_triples)
Completed triple $\leftarrow \mathbf{LLM}(\mathcal{P}, \mathcal{T}')$
**return** Completed triple

The core idea of the retrieval-augmented generation technique is to dynamically retrieve relevant external information while using generative pre-trained language models, thereby supporting and grounding the generated content. The implementation of RAG typically involves two key components: 1) the retrieval module, which is responsible for retrieving the most relevant information from a pre-constructed text corpus or vector database based on the current input. The retrieval module can leverage traditional information retrieval techniques, such as inverted indices, or more advanced vector-based similarity search techniques; 2) the generation model, which receives the relevant information retrieved by the retrieval module along with the original input and synthesizes this information to generate the target output.

Next, we introduce the implementation architecture of the knowledge graph completion model based on retrieval-augmented generation technology, which we have named RA-KG-LLM. This architecture is divided into two main stages: the preparation stage and the generation stage. The goal is to enhance the efficiency and accuracy of knowledge graph completion by combining the generative capabilities of large models with the retrieval augmentation mechanism. The schematic diagram of the architecture is shown in Fig. 1. The overall algorithmic process of the model is presented in Algorithm 1.

In the preparation phase, the first step involves vectorizing the triples in the knowledge graph to facilitate efficient similarity retrieval later in the vector database. The objective of vectorization is to transform each triple $\mathcal{T}$ into a point in high-dimensional space, that is, a vector $\mathbf{v}$, represented by the embedding function $f$ as

$$\mathbf{v} = f(\mathcal{T}) = f(h, r, t).$$

This ensures that similar triples are closer in vector space. To accommodate knowledge graphs in different languages, we have adopted language-specific embedding models: the bge-large-zh model [20] for vectorizing triples in Chinese knowledge graphs and the bge-large-en model [21] for English knowledge graphs. Both models are based on large-scale pre-trained text embedding techniques, which effectively capture the semantic information within the triples. Specifically, the embedding models capture semantic information by maximizing the similarity between the input text and its context, with the pre-training objective expressed as

$$\mathcal{L} = -\sum_{i=1}^{n} \log P(x_i \mid x_{<i}, x_{>i}),$$

where $\mathcal{L}$ is the loss function, and $P(x_i \mid x_{<i}, x_{>i})$ represents the probability of the word $x_i$ occurring given the context. The vectorized triples are stored in a vector database based on FAISS (Facebook AI Similarity Search) [22]. FAISS is an efficient similarity search library designed for processing and retrieving large-scale vector data. It utilizes optimized data structures and algorithms such as Product Quantization and Hierarchical Clustering to achieve fast and accurate similarity searches on large datasets. By storing the vectorized triples in the FAISS database, we can quickly retrieve the triples most similar to the current query when entity and relationship completion is needed, thereby enhancing the completion efficiency and accuracy of the knowledge graph.

In the generation stage, retrieval-augmented generation technology is employed to perform the task of knowledge graph completion. Specifically, when given a knowledge graph triple $\mathcal{T}'$ that needs completion or prediction, it is first vectorized, represented as vector $\mathbf{v}'$.This vector is then used as a query for similarity retrieval in the vector database. The goal of similarity retrieval is to find the $k$ most similar vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$, whose corresponding triples are considered most relevant to the current completion task.

To achieve efficient similarity retrieval, we utilize the FAISS (Facebook AI Similarity Search) library. Within FAISS, product quantization [23] is a common vector quantization method that significantly reduces storage space and computational complexity by decomposing high-dimensional vectors into multiple low-dimensional sub-vectors and quantizing them separately. The formula for product quantization can be expressed as:

$$\mathbf{v} \approx [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m] \cdot [\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_m]$$

where $\mathbf{v}_i$ are the sub-vectors and $\mathbf{c}_i$ are the corresponding quantization centers.Furthermore, FAISS also employs a hierarchical quantization algorithm [24] , which accelerates similarity searches by constructing a hierarchical structure of vectors. The basic idea of hierarchical quantization is to

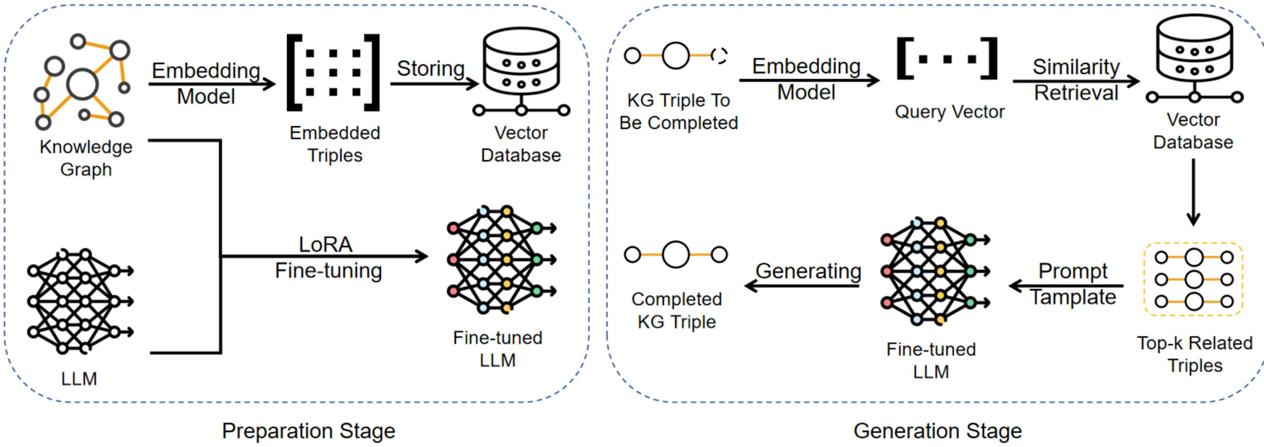



Fig. 1. Architecture of RA-KG-LLM (Knowledge Graph Completion Model Based on Retrieval-Augmented Generation)

build a tiered nearest neighbor graph. In this graph, the top layer contains a few central vectors, while the bottom layer contains all vectors. The search process starts at the top layer and proceeds downwards through precise matching, rapidly narrowing the search scope to find the most similar vectors.

The $k$ most similar triples retrieved are then integrated into prompt words, which are used to guide a large model in performing the knowledge graph completion task. The design of the prompt is based on the assumption that knowledge information similar to the triple to be completed can provide valuable contextual information for the generation model, thereby helping the model more accurately predict missing entities or relationships.

Specifically, the prompt are formed by taking the retrieved relevant triples $\{(s_1, p_1, o_1), (s_2, p_2, o_2), \ldots, (s_k, p_k, o_k)\}$ as input, combined with the current triple $\mathcal{T}'$ to be completed, creating an augmented context. Upon receiving these prompt, the generation model leverages its powerful generative capabilities to predict the most likely missing entity or relationship, thus completing the knowledge graph completion task.

We select Baichuan2-13B[25] as the large model for our RA-KG-LLM due to its superior performance and ability to capture complex semantic information. Furthermore, we employ the Low-Rank Adaptation [26] (LoRA) technique to effectively inject semantic information from the knowledge graph into the model. LoRA is a parameter-efficient fine-tuning method that introduces low-rank matrices into specific layers of the pre-trained model, thereby significantly reducing the number of parameters that need to be updated during fine-tuning.

Specifically, let the weight matrix of the pre-trained model be $\mathbf{W} \in \mathbb{R}^{d \times d}$. During LoRA fine-tuning, we introduce two low-rank matrices $\mathbf{A} \in \mathbb{R}^{d \times r}$ and $\mathbf{B} \in \mathbb{R}^{r \times d}$, where $r \ll d$. The fine-tuned weight matrix can be expressed as:

$$\mathbf{W}' = \mathbf{W} + \mathbf{AB}$$

where $\mathbf{A}$ and $\mathbf{B}$ are the parameters to be learned during fine-tuning. This approach allows us to inject new semantic information while preserving the original capabilities of the model.In the final generation task, the large model leverages not only its built-in semantic knowledge and generative capabilities but also the external knowledge retrieved, enhancing the accuracy of its predictions. This approach effectively utilizes the semantic information within the knowledge graph and the relational information among the triples while maintaining good generalizability and scalability.

we will present the specific implementation details in the experimental analysis, including comparative experiments conducted on several general and legal domain datasets to validate the effectiveness of the knowledge graph completion method based on retrieval-augmented generation with large models.

## IV. EXPERIMENT

### A. Experiment Settings

Our experiments encompass several well-known general domain datasets, including CoDeX-S[27], FB15K-237N[28], WN18RR[29], and YAGO3-10[30]. Additionally, we introduce a legal domain dataset, Cail2022[31], which includes a substantial amount of specialized terminology and complex semantic relationships, thus increasing the difficulty of knowledge graph completion.

The baseline methods for comparison in our experiments include the following approaches:

- KG-BERT[4], which models knowledge graph completion as a binary text classification task using a pre-trained language model
- KGT5[12], a sequence-to-sequence method based on the encoder-decoder pre-trained language model T5
- KG-Llama[18], which employs fine-tuning of a large model for knowledge graph completion
- KoPA[19], which integrates structural embedding models with the capabilities of large models.

We selected four representative large models for our experiments: Alpaca[32], Llama3-8B[33], Chatglm3-6b[34], and

Baichuan2-13B[25]. We use the Baichuan2-13b model as the base model for our RA-KG-LLM due to its larger scale and superior performance in capturing complex semantic information. The other three models are used for ablation studies to compare and contrast their effectiveness with Baichuan2-13B. These models have demonstrated outstanding performance in recent AI research and are widely applied in various natural language processing tasks.

Our experiments include three types of tasks:

- Triple prediction, where the model needs to judge the correctness of a triple
- Entity prediction, where the model needs to predict missing entities
- Relationship prediction, where the model needs to predict missing relationships

For the triple prediction task, we selected F1 score as the evaluation metric; for entity prediction and relation prediction tasks, we employed Hits@1 as the evaluation metric, which is the probability that the top-ranked entity predicted by the model is the target entity.

Furthermore, we designed ablation experiments to compare the following four scenarios:

- Direct use of a large model
- Large models combined with retrieval-augmented generation techniques without fine-tuning
- Fine-tuned large model
- Fine-tuned large models integrated with retrieval-augmented generation techniques

### B. Triple Prediction Experiments

Tables II and III display the effectiveness of fine-tuned large models integrated with retrieval-augmented generation techniques in the triple prediction task on general and legal domain datasets respectively, measured by F1 scores. The best results in the baselines and our methods are underscored for emphasis.

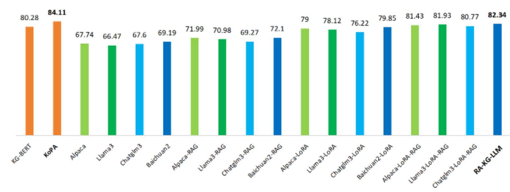
TABLE II
THE PERFORMANCE OF FINE-TUNED LARGE MODELS INCORPORATING RETRIEVAL-AUGMENTED GENERATION TECHNIQUES IN TRIPLE PREDICTION TASKS ON GENERAL DATASETS (F1 SCORE)

| Method | CoDeX-S | FB15K-237N |
|---|---|---|
| KG-BERT | 80.28 | 67.84 |
| KoPA | 84.11 | 80.81 |
| Alpaca-LoRA-RAG | 81.43 | 78.09 |
| Llama3-LoRA-RAG | 81.93 | 78.41 |
| Chatglm3-LoRA-RAG | 80.77 | 77.47 |
| RA-KG-LLM | **82.34** | **79.28** |

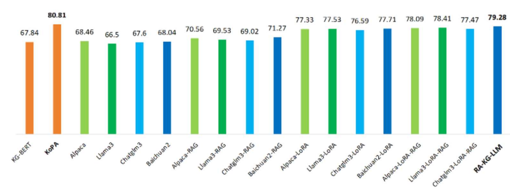
Fig. 2 visually presents the ablation comparisons of our methods for the triple prediction task on general and legal domain datasets, respectively, also measured by F1 scores. Here, 'Alpaca' denotes the direct use of large models; 'Baichuan-RAG' indicates large models combined with retrieval-augmented generation techniques without fine-tuning; 'Alpaca-LoRA' refers to fine-tuned large models; and 'Alpaca-LoRA-RAG' signifies fine-tuned large models integrated with retrieval-augmented generation techniques.

TABLE III
THE PERFORMANCE OF FINE-TUNED LARGE MODELS INCORPORATING RETRIEVAL-AUGMENTED GENERATION TECHNIQUES IN TRIPLE PREDICTION TASKS ON LEGAL DATASETS (F1 SCORE)

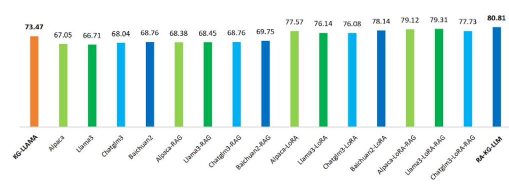
| Method | P | R | F1 |
|---|---|---|---|
| KG-Llama | 80.28 | 67.84 | 73.47 |
| Alpaca-LoRA-RAG | 80.23 | 78.04 | 79.12 |
| Llama3-LoRA-RAG | 79.99 | 78.64 | 79.31 |
| Chatglm3-LoRA-RAG | 78.54 | 76.95 | 77.73 |
| RA-KG-LLM | 81.63 | 80.01 | **80.81** |




(a) Ablation experiments for triple prediction on CoDeX-S




(b) Ablation experiments for triple prediction on FB15K-237N




(c) Ablation experiments for triple prediction on Cail2022

Fig. 2. Bar chart of ablation experiments for triple prediction tasks (F1 Score)

The experiments validate the efficacy of knowledge graph triple prediction models based on large models enhanced by retrieval generation in both general and legal domains. Table II shows specific results on general datasets, where all four large model configurations surpassing the KG-BERT baseline, with RA-KG-LLM reaching an F1 of 82.34 on CoDeX-S and 79.28 on FB15K-237N. Although there is still a gap of about 2% compared to the specialized KoPA method, considering the versatility of our proposed approach for various tasks, this gap is acceptable. Table III displays results on a legal dataset, where our knowledge graph triple prediction model RA-KG-LLM surpasses the KG-Llama baseline, achieving an F1 of 80.81, confirming the effectiveness of our method in legal domain knowledge graph completion tasks.

The ablation analysis, as shown in Fig. 2, reveals similar trends across both general and legal datasets. When large models are directly applied to knowledge graph triple prediction tasks, performance is poor; for instance, Alpaca achieves only a 67.74 F1 on CoDeX-S, as the model tends to validate all input triples as true, which is nearly unusable in practical applications. When large models without fine-tuning are applied together with retrieval-augmented generation techniques, although the model still tends to validate input triples as true, the retrieval framework provides relevant triple references that enable correct predictions in some cases, with an average F1 increase of 1.195 across four foundational models on the legal dataset. The Baichuan2-13b model, due to its larger parameter size, shows superior performance over other large models, achieving a 72.1 F1 on CoDeX-S. Fine-tuned large models significantly improve performance in the knowledge graph triple prediction task, with Baichuan2-13B improving by 14.21% on CoDeX-S compared to before fine-tuning, and an average improvement of 13.8% across all four foundational models on the legal dataset. This indicates that large models, through fine-tuning, gain a deeper understanding of the semantic information of knowledge graphs and handle triple prediction tasks more accurately. The ablation experiments confirm the effectiveness of each module in the knowledge graph triple prediction model based on large models with retrieval-augmented generation and the rationality of the model design.

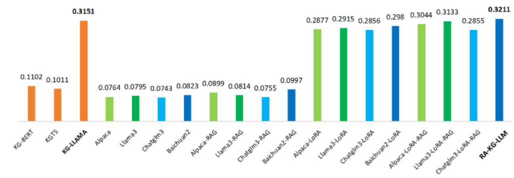
### C. Entity Prediction Experiments

Table IV displays the performance of fine-tuned large models integrated with retrieval-augmented generation techniques for entity prediction tasks on general and domain-specific datasets, measured by Hits@1 values, where the best results in the baselines and our methods are underscored for emphasis.

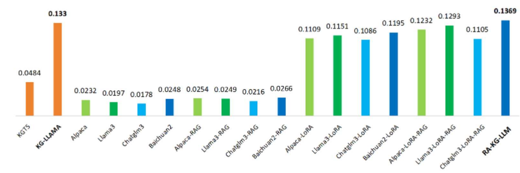
TABLE IV
THE PERFORMANCE OF FINE-TUNED LARGE MODELS INCORPORATING RETRIEVAL-AUGMENTED GENERATION TECHNIQUES IN ENTITY PREDICTION TASKS (HITS@1)

| Method | WN18RR | YAGO3-10 | Cail2022 |
|---|---|---|---|
| KG-BERT | 0.1102 | - | - |
| KGT5 | 0.1011 | 0.0484 | - |
| KG-Llama | 0.3151 | 0.1330 | 0.2745 |
| Alpaca-LoRA-RAG | 0.3044 | 0.1232 | 0.2702 |
| Llama3-LoRA-RAG | 0.3133 | 0.1293 | 0.2736 |
| Chatglm3-LoRA-RAG | 0.2855 | 0.1105 | 0.2659 |
| RA-KG-LLM | **0.3211** | **0.1369** | **0.2861** |

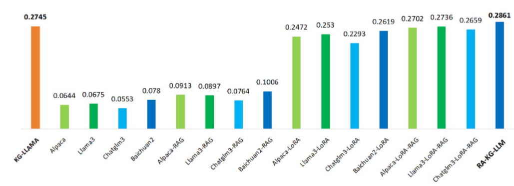
Fig. 3 visually compares the ablation comparisons of our methods on entity prediction tasks across general and domain-specific datasets, also measured by Hits@1 values.As in previous experiments, 'Alpaca' represents the direct use of large models; 'Alpaca-LoRA' denotes large models only fine-tuned on general datasets using LoRA; 'Baichuan-RAG' indicates large models combined with retrieval-augmented generation techniques without fine-tuning; and 'Alpaca-LoRA-RAG' signifies fine-tuned large models integrated with retrieval-augmented generation techniques.




(a) Ablation experiments for entity prediction on WN18RR




(b) Ablation experiments for entity prediction on YAGO3-10




(c) Ablation experiments for entity prediction on Cail2022

Fig. 3. Bar chart of ablation experiments for entity prediction tasks (Hits@1)

The knowledge graph completion models based on large models with retrieval-augmented generation prove effective for entity prediction tasks in both general and legal domains. As shown in Table IV, the knowledge graph entity prediction models enhanced by large models with retrieval-augmented generation surpass baseline methods on three datasets within the legal domain; for example, RA-KG-LLM achieves a Hits@1 of 0.2861 on the Cail2022 dataset, which is a 4.23% improvement over the baseline KG-Llama. These results validate the effectiveness of our proposed method for entity prediction tasks in general and legal domain knowledge graphs.

The ablation analysis, as illustrated in Fig. 3, shows that when large models are directly used, they primarily rely on the inherent knowledge learned during the pre-training phase to predict missing entities, leading to relatively lower performance, with an average Hits@1 of only 0.0663 on the legal domain dataset Cail2022. This outcome indicates that relying solely on the inherent knowledge of large models is insufficient for complex knowledge graph entity prediction tasks. When large models without fine-tuning are applied with retrieval-augmented generation techniques, the performance is moderately improved due to the reference of relevant triples provided by the retrieval framework, with an average Hits@1 improvement rate of 35% on the legal domain dataset Cail2022. Fine-tuned large models show a significant improvement in performance on knowledge graph entity prediction tasks, with an average Hits@1 increase of 0.7262 on the legal domain dataset Cail2022. This suggests that the fine-tuning process effectively enhances the model's understanding of semantic information in the knowledge graph, thereby improving its predictive capabilities. The ablation experiments confirm the effectiveness and rational design of each module in the knowledge graph completion model based on large models with retrieval-augmented generation for entity prediction tasks in both general and legal domains.

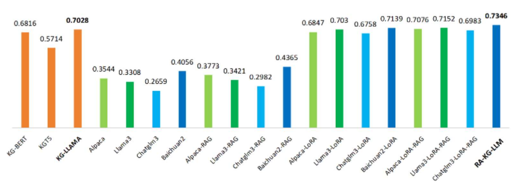
### *D. Relation Prediction Experiments*

Table V demonstrates the performance of fine-tuned large models integrated with retrieval-augmented generation techniques for relation prediction tasks on general and domain-specific datasets, measured by Hits@1 values, with the best results in the baselines and our methods underscored for emphasis.

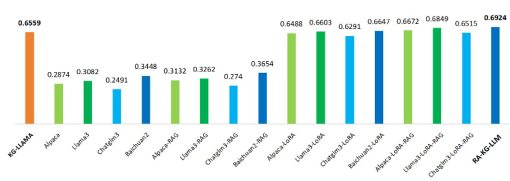
TABLE V
THE PERFORMANCE OF FINE-TUNED LARGE MODELS INCORPORATING RETRIEVAL-AUGMENTED GENERATION TECHNIQUES IN RELATION PREDICTION TASKS (HITS@1)

| Method | YAGO3-10 | Cail2022 |
|---|---|---|
| KG-BERT | 0.6816 | - |
| KGT5 | 0.5714 | - |
| KG-Llama | 0.7028 | 0.6559 |
| Alpaca-LoRA-RAG | 0.7076 | 0.6672 |
| Llama3-LoRA-RAG | 0.7152 | 0.6849 |
| Chatglm3-LoRA-RAG | 0.6983 | 0.6515 |
| RA-KG-LLM | **0.7346** | **0.6924** |

Fig. 4 visually presents the ablation comparison of different methods for relation prediction tasks across general and domain-specific datasets, also measured by Hits@1 values.




(a) Ablation experiments for relation prediction on YAGO3-10




(b) Ablation experiments for relation prediction on Cail2022

Fig. 4. Bar chart of ablation experiments for relation prediction tasks (Hits@1)

The experiments validate the effectiveness of knowledge graph triple prediction models based on large models enhanced by retrieval generation for relation prediction tasks in both general and legal domains. As indicated in Table V, the knowledge graph relation prediction models utilizing large models with retrieval-augmented generation exceed the baseline methods on three datasets within the legal domain; for instance, RA-KG-LLM achieves a Hits@1 of 0.6924 on a legal domain dataset, which is a 5.56% improvement over the baseline KG-Llama. Furthermore, it is observed that the retrieval-augmented methods based on the larger parameter foundation of Baichuan2-13b consistently achieve the best results.

The ablation analysis, as shown in Fig. 4, reveals that when large models are directly used, they primarily rely on the inherent knowledge acquired during the pre-training phase to predict missing relations, resulting in poor performance both in general and legal domain datasets. For example, the average Hits@1 on the legal domain dataset Cail2022 is only 0.2973, significantly lower than the baseline. This indicates that relying solely on the original large models has clear limitations in knowledge graph relation prediction tasks within the legal domain. When retrieval-augmented generation techniques are applied, due to the reference of relevant triples provided by the retrieval framework, the performance is moderately improved, with an average Hits@1 improvement rate of 7.5% on the legal domain dataset Cail2022. Fine-tuned large models show a significant improvement in performance on knowledge graph relation prediction tasks, effectively incorporating semantic information from the knowledge graph through fine-tuning, thereby enhancing their predictive capabilities. On the legal domain dataset Cail2022, the average Hits@1 increases by 0.3534. The ablation experiments confirm the effectiveness and rational design of each module in the knowledge graph completion model based on large models with retrieval-augmented generation for relation prediction tasks in both general and legal domains.

Through a series of experiments, we comprehensively evaluated the performance of our proposed method in knowledge graph triple prediction, entity prediction, and relation prediction tasks. These experiments covered not only general domain datasets such as WN18RR and YAGO3-10 but also specialized domain datasets like Cail2022, validating the versatility and effectiveness of the knowledge graph completion models based on large models with retrieval-augmented generation in both general and legal domains.

## V. CONCLUSION

In response to the numerous challenges faced in the task of completing knowledge graphs within the legal domain—such as vast and specialized domain knowledge, intricate entity relationships, and high demands for inferential accuracy—we propose a model for the completion of entities and relationships in knowledge graphs based on Retrieval-Augmented Generation with large models, which we have named RA-KG-LLM. This model leverages the generative capabilities of large models combined with the recently developed Retrieval-Augmented Generation (RAG) framework, balancing the semantic and associative information of knowledge graphs to effectively enhance the performance of knowledge graph completion tasks.

We have conducted experimental validations on multiple general-purpose datasets in both Chinese and English, as well as on the specialized legal dataset cail2022, covering three main tasks: triple prediction, entity prediction, and relationship prediction. The methods proposed in this study achieved performance surpassing baseline levels across multiple datasets. Our research demonstrates the potential applications of large model retrieval-augmented generation technology in knowledge graph reasoning, offering new approaches to addressing the challenges in legal domain knowledge graph completion.

In the future, we plan to further explore the applications of large model technologies in the field of knowledge graphs, such as the integration of multimodal large models with cross-modal knowledge graphs, and the study of the interpretability of knowledge graph completion methods based on large models.

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